

Katie Warburton · Computational Cognitive Scientist

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Education

Ph.D. in Cognitive Science

2021 - present

University of Melbourne; University of Toronto

Melbourne, Australia; Toronto, Canada

- **Supervisors:** Charles Kemp, Yang Xu, Lea Frermann
- Jointly awarded degree from the UofT-UoM International Research Training Group

B.Comp (Hons) with a Specialization in Cognitive Science

2017 - 2021

Queen's University

Kingston, Canada

- Received the medal in computing which is awarded to the student who has achieved the highest standing in a plan offered by the School of Computing.

Research Experience

UofT-UoM International Research Training Group

Sept. 2021 - Present

Doctoral Candidate

- **Thesis:** A Computational Analysis of the Structure and Evolution of Hierarchical Category Systems
- **Supervisors:** Charles Kemp, Yang Xu, Lea Frermann
- Previously developed large-scale computational methods for quantifying biases in hierarchical category systems, using library classification systems and subject headings as case studies
- Currently combining computational and experimental methods to explore sequence effects in the evolution of hierarchical category systems

Queen's School of Computing

May 2020 - April 2021

Undergraduate Thesis

- **Title:** Using Correlations of Word Frequency to Find Clusters in Unstructured Radiology Reports
- **Supervisor:** Randy Ellis
- Researched and implemented methods for the natural language processing of lung cancer radiology reports
- Applied topic modeling and clustering algorithms to aid in report processing
- Funded by an NSERC Undergraduate Student Research Award

Dept. of Languages, Literatures and Cultures, Queen's University

June 2019 - Aug. 2019

Undergraduate Research Assistant

- **Supervisors:** Nathan Thanyehténhas Brinklow, Lorena Jessop
- Assisted with the development of an automatic speech recognition model for Kanyen'kéha, a polysynthetic language

Publications

Warburton, K., Frermann, L., Xu, Y. & Kemp, C. (Under Review). An Experimental Study of the Evolution of Hierarchical Category Systems.

Warburton, K., Kemp, C. & Frermann, L. (2025). Scientists & Women Scientists: Exploring Gender Biases in Institutional Category Systems. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 47.

Warburton, K., Kemp, C., Xu, Y. & Frermann, L. (2024). Quantifying Bias in Hierarchical Category Systems. *Open Mind: Discoveries in Cognitive Science*, 8, 102-130.

Warburton, K., Kemp, C., Xu, Y. & Frermann, L. (2023). Quantifying Bias in Library Classification Systems. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 45.

Presentations

- Warburton, K.** (March, 2026). *A Computational Analysis of the Structure and Evolution of Hierarchical Category Systems*. Complex Human Data Hub Seminar Series, Melbourne, Australia. **Research Seminar.**
- Warburton, K.,** Kemp, C., & Frermann, L. (July, 2025). *Scientists & Women Scientists: Exploring Gender Biases in Institutional Category Systems*. Annual Meeting of the Cognitive Science Society, San Francisco CA, United States. **Conference Poster.**
- Warburton, K.** (July, 2025). *Categorization in Practice: Insights from Library Classification Systems*. Lupyran Lab, Madison WI, United States. **Invited Talk.**
- Warburton, K.,** Kemp, C., Xu, Y. & Frermann, L. (July, 2023). *Quantifying Bias in Library Classification Systems*. Annual Meeting of the Cognitive Science Society, Sydney, Australia. **Conference Poster.**

Teaching Experience

- University of Melbourne** Melbourne, Australia
Academic Tutor 2024
• **PSYC30013:** Research Methods for Human Inquiry (Semester 1 2024)
- University of Toronto** Toronto, Canada
Guest Speaker November 2022
• **Title:** Quantifying Biases in Library Classification Systems
• **Course:** Data, Computation, and The Mind (COG260).
- University of Toronto** Toronto, Canada
Teaching Assistant 2021-2022
• **CSC199:** Intelligence, Artificial and Human (Fall Term 2022)
• **CSC2611:** Computational Models of Semantic Change (Fall Term 2022)
• **CSC110:** Foundations of Computer Science (Fall Term 2021)
- Queen's University** Kingston, Canada
Teaching Assistant 2018-2021
• **CISC352:** Artificial Intelligence (Winter Term 2021)
• **COGS201:** Cognition and Computation (Fall Term 2019; 2020)
• **COGS100:** Introduction to Cognitive Science (Summer Term 2019; Winter Term 2020)
• **CISC121:** Introduction to Computer Science (Fall Term 2018)

Awards

- 2025 **Career Development Award**, Melbourne School of Psychological Sciences
- 2021- 2026 **Melbourne Research Scholarship**, University of Melbourne
- 2021 **Medal in Computing**, Queen's School of Computing
- 2020 **NSERC Undergraduate Student Research Award**, Queen's University
- 2018 **Queen's Appeal Undergraduate Scholarship**, Queen's University
- 2017 **Principal's Scholarship**, Queen's University

Outreach & Professional Development

SERVICE & OUTREACH

- 2024 - 2025 **Graduate Researchers in Psychological Sciences (GRiPS)**, President University of Melbourne
- 2023 **Science Rendevous UofT**, Volunteer University of Toronto
- 2019 - 2020 **Computing Students Association**, Governance & Internal Affairs Director Queen's University

PEER REVIEW

Annual Meeting of the Cognitive Science Society

Skills

Programming Languages: Python, Javascript, R, MATLAB, Prolog, Java, Haskell.

Other: $\text{\LaTeX}$, HTML/CSS, SQL, and Markdown.

References

Available upon request